**Definition 1.** This is a package for theorem environments.

# theofig

*figure implementation of theorem environments*

A Typst package for creation and customization
of theorem environments built on top of `std.figure`.

github.com/Danila-Bain/typst-theorems

**Version 0.2.0**

## Guide


Usage examples . . . 2
- Basic usage . . . 2
- Default environments . . . 2
- Custom environments . . . 2
- Languages support . . . 3
- Ways to specify numbering style . . . 3
- Numbers out of order . . . 3
- Shared numbering . . . 4
- Show rules to specify a style . . . 5
- Style examples . . . 5
- Page breaking . . . 6
- Limitations . . . 8


## Reference


Main functions . . . 9
- theofig() . . . 9
- show-figure-caption() . . . 14
- figure-where-kind-in() . . . 14
- theofig-reset-counters() . . . 15
- algorithm() . . . 15
- corollary() . . . 15
- definition() . . . 15
- example() . . . 16
- lemma() . . . 16
- problem() . . . 16
- proof() . . . 16
- proposition() . . . 16
- remark() . . . 16
- solution() . . . 16
- statement() . . . 16
- theorem() . . . 16
- theofig-kinds . . . 16
- theofig-translations . . . 17
- theofig-split-box() . . . 17

# Usage examples

```
#import "@preview/theofig:0.2.0": *
```

## Basic usage

```
#theorem[#lorem(5)] <theorem-1>

#theorem[Lorem][#lorem(10)]

#proof[It follows from @theorem-1.]
```

**Theorem 1.** Lorem ipsum dolor sit amet.

**Theorem 2 (Lorem).** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

**Proof.** It follows from Theorem 1. ∎

## Default environments

`theofig` defines a number of default environments with sensible default style.

```
#theorem[#lorem(5)]

#lemma[#lorem(5)]

#statement[#lorem(5)]

#remark[#lorem(5)]

#corollary[#lorem(5)]

#example[#lorem(5)]

#definition[#lorem(5)]

#algorithm[#lorem(5)]

#proof[#lorem(5)]

#problem[#lorem(5)]

#solution[#lorem(5)]
```

**Theorem 1.** Lorem ipsum dolor sit amet.

**Lemma 1.** Lorem ipsum dolor sit amet.

**Statement 1.** Lorem ipsum dolor sit amet.

**Remark 1.** Lorem ipsum dolor sit amet.

**Corollary.** Lorem ipsum dolor sit amet.

**Example 1.** Lorem ipsum dolor sit amet.

**Definition 1.** Lorem ipsum dolor sit amet.

**Algorithm 1.** Lorem ipsum dolor sit amet.

**Proof.** Lorem ipsum dolor sit amet. ∎

**Problem 1.** Lorem ipsum dolor sit amet.

**Solution.** Lorem ipsum dolor sit amet.

## Custom environments

All default environments of `theofig` package are defined as `.with()`-specializations of a function `theofig()`, which can also be used to create custom environments.

```
#let joke = theofig.with(supplement: "Joke")

#{theofig-kinds += ("joke", )}

#show figure-where-kind-in(theofig-kinds): set
text(gradient.linear(red, green, blue))

#joke[Why was six afraid of seven? Because 7 8 9.]
#joke[
  Parallel lines have so much in common...
  It's a shame they'll never meet.
]
#statement[A topologist is someone who can't tell
the difference between a coffee mug and a
doughnut.]
```

**Joke 1.** Why was six afraid of seven? Because 7 8 9.

**Joke 2.** Parallel lines have so much in common... It's a shame they'll never meet.

**Statement 1.** A topologist is someone who can't tell the difference between a coffee mug and a doughnut.

If the `supplement` option of `theofig()` is specified, the `kind` is chosen automatically as `lower(supplement)`, so in this example adding `"joke"` to `theofig-kinds` lets us apply styling to a `joke` environment together with all standard environments.

## Languages support

Enabled by the `translate-supplement` option of `theofig()`, `supplement` is translated based on context-dependant `text.lang`. List of supported languages: cs, de, en, es, fr, it, ja, ko, pl, pt, ru, zh. Note that unlike supplement of a figure, a supplement in reference changes if `text.lang` is not the same as it was at the location of the figure.

```
#set text(lang: "ru")
#theorem[#lorem(5)]

#set text(lang: "es")
#theorem[#lorem(5)]

#set text(lang: "de")
#theorem[#lorem(5)]

#set text(lang: "ja")
#theorem[#lorem(5)]
```

**Теорема 1.** Lorem ipsum dolor sit amet.

**Teorema 2.** Lorem ipsum dolor sit amet.

**Satz 3.** Lorem ipsum dolor sit amet.

**定理 4.** Lorem ipsum dolor sit amet.

## Ways to specify numbering style

Most default environments use `figure`'s default numbering, which is `"1"` (see Definition 1). Hence, we can change default numbering for many environments simultaneously using a `show` rule with `set figure(numbering: ...)` (see Definition II). Then, if we specify an argument `numbering` in the environment, it takes priority over `figure`'s numbering (see Definition C and Definition (iv)).

```
#definition[Default @def-a-1.]<def-a-1>

#show figure-where-kind-in(
  theofig-kinds
): set figure(numbering: "I")
#definition[Show rule @def-a-2.]<def-a-2>

#let definition = definition.with(numbering: "A")
#definition[Redefined @def-a-3.]<def-a-3>

#definition(numbering: numbering.with("(i)"))[
  Argument @def-a-4.
]<def-a-4>
```

**Definition 1.** Default Definition 1.

**Definition II.** Show rule Definition II.

**Definition C.** Redefined Definition C.

**Definition (iv).** Argument Definition (iv).

## Numbers out of order

Using the `number` option of `theofig()`, we can specify a number regardless of automatic numeration. Passing a `label` to `number` copies the `numbering` of the same environment by that `label`, which is useful for alternative definitions or equivalent statements of theorems (see Definition 2 and Definition 2').

```
#definition[Default.]

#definition(numbering: none)[No numbering.]

#definition[Equivalent to @def-2.]<def-1>

#definition(number: <def-1>, numbering: "1'")[
  Equivalent to @def-1.
]<def-2>

#definition(number: 100)[
  This is @def-100.
]<def-100>

#definition(number: 5, numbering: "A")[
  This is @def-3.
]<def-3>

#definition(number: $e^pi$)[
  This is @def-exp
]<def-exp>

#definition[Back to default.]
```

**Definition 1.** Default.

**Definition.** No numbering.

**Definition 2.** Equivalent to Definition 2'.

**Definition 2'.** Equivalent to Definition 2.

**Definition 100.** This is Definition 100.

**Definition E.** This is Definition E.

**Definition $e^\pi$.** This is Definition $e^\pi$

**Definition 3.** Back to default.

Another use case for `number` argument is local numbering, such as multiple corollaries immediately after a theorem:

```
#theorem[]
#corollary[]
#theorem[]
#corollary(number: "1")[]
#corollary(number: "2")[]
#theorem[]
#corollary(number: "1")[]
#corollary(number: "2")[]
```

**Theorem 1.**

**Corollary.**

**Theorem 2.**

**Corollary 1.**

**Corollary 2.**

**Theorem 3.**

**Corollary 1.**

**Corollary 2.**

## Shared numbering

If you want different environments to share numbering, you just need to have them have the same `kind`, but different `supplement`:

```
#let lemma     = lemma.with(kind: "theorem")
#let statement = statement.with(kind: "theorem")
#show figure.where(kind: "theorem"): it => {
    show figure.caption: text.with(blue); it
  }
#theorem[#lorem(5)]
#lemma[#lorem(5)]
#statement[#lorem(5)]
#theorem[#lorem(5)]
```

**Theorem 1.** Lorem ipsum dolor sit amet.

**Lemma 2.** Lorem ipsum dolor sit amet.

**Statement 3.** Lorem ipsum dolor sit amet.

**Theorem 4.** Lorem ipsum dolor sit amet.

One obvious limitation of that approach is that not only numbering will be shared. All styling of `theorem()` that is based on `show` rules will also apply to `lemma()` and `statement()`. To mitigate that, styling can be applied individually through setting arguments `format-caption`, `format-body`, `block-options`, and `figure-options`.

```
#show figure.where(kind: "theorem"): it => {
    show figure.caption: text.with(blue); it
  }
#let theorem = theorem.with(
  kind: "theorem",
  block-options: (
    stroke: 1pt, radius: 3pt, inset: 5pt,
  ),
)
#let lemma     = lemma.with(
  kind: "theorem",
  format-caption: none,
)
#let statement = statement.with(
  kind: "theorem",
  format-body: emph,
)

#theorem[#lorem(5)]
#lemma[#lorem(5)]
#statement[#lorem(5)]
```

**Theorem 1.** Lorem ipsum dolor sit amet.

Lemma 2. Lorem ipsum dolor sit amet.

**Statement 3.** *Lorem ipsum dolor sit amet.*

## Show rules to specify a style

All environments are `figure`'s under the hood, and they can be styled using show rules. The title of environment, such as "**Theorem 4 (Cauchy).**" can be styled using show rules for `figure.caption`.

```
// apply to one
#show figure.where(kind: "theorem"): smallcaps
// apply to some
#show figure-where-kind-in(
  ("solution", "problem")
): emph
// apply to all
#show figure-where-kind-in(theofig-kinds): set
figure(
  numbering: "I",
)
// apply to all except some
#show figure-where-kind-in(
  theofig-kinds, except: ("proof",),
): set text(blue)

#definition[#lorem(10)]
#theorem[#lorem(10)]
#proof[#lorem(10)]
#problem[#lorem(10)]
#solution[#lorem(10)]
```

**Definition I.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

**Theorem I.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

**Proof.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do. ∎

***Problem I.*** *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.*

***Solution.*** *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.*

## Style examples

Note that in the following examples, in order for `block`-ed styles to be breakable without visual glitches, you should make blocks inside a `figure` sticky with something like
`show figure-where-kind-in(theofig-kinds): set block(breakable: true, sticky: true)`.

```
#theorem[Default. #lorem(16)]

#show figure.where(kind: "definition"): it => {
  show figure.caption: emph
  show figure.caption: strong.with(delta: -300)
  it
}
#definition[Italic caption. #lorem(16)]

#show figure.where(kind: "lemma"): it => {
  show figure.caption: underline.with(offset:
1.5pt)
  show figure.caption: strong.with(delta: -300)
  it
}
#lemma[Underline caption. #lorem(16)]

#show figure.where(kind: "proposition"): it => {
  show: emph
  show figure.caption: emph
  show figure.caption: smallcaps
  show figure.caption: strong.with(delta: -300)
  it
}
#proposition[Italic body, smallcaps caption.
#lorem(12)]


#show figure.where(kind: "corollary"): it => {
  show figure.caption: strong.with(delta: -300)
  show figure.caption: set text(tracking: 3pt)
  it
}
#corollary[Sparse caption. #lorem(16)]

#show figure.where(kind: "statement"): block.with(
  stroke: 1pt, radius: 3pt, inset: 5pt,
)
#statement[Block. #lorem(16)]

#show figure.where(kind: "solution"): block.with(
  stroke: (left: 1pt), inset: (right: 0pt, rest:
5pt)
)
#solution[Line to the left. #lorem(16)]
```

**Theorem 1.** Default. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et.

*Definition 1.* Italic caption. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et.

Lemma 1. Underline caption. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et.

PROPOSITION 1. *Italic body, smallcaps caption. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor.*

C o r o l l a r y. Sparse caption. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et.

**Statement 1.** Block. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et.

**Solution.** Line to the left. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et.

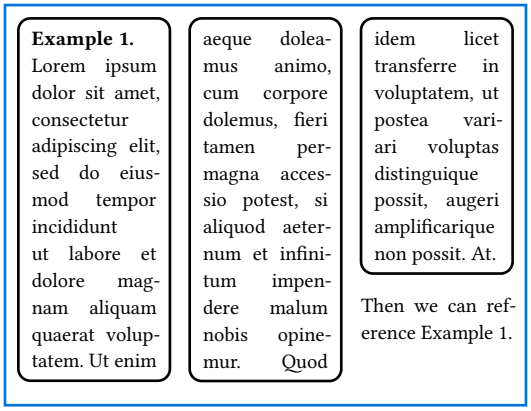
## Page breaking

To enable page breaking of theorem environments, one must add

```
#show figure-where-kind-in(theofig-kinds): set block(breakable: true)
```

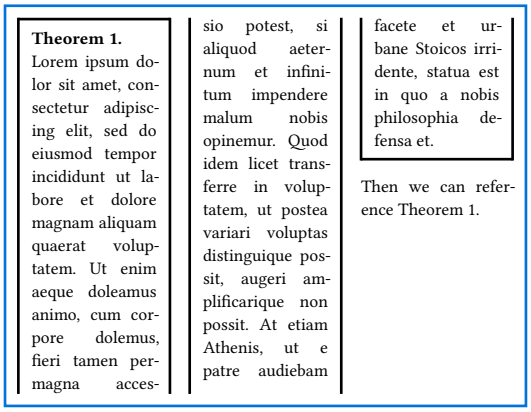
line to the preambule. This show rule is set by user and not done internally because setting block options of figure is impossible without a show rule, and doing this before labels are attached to the figures leads to “cannot reference styled” issues.

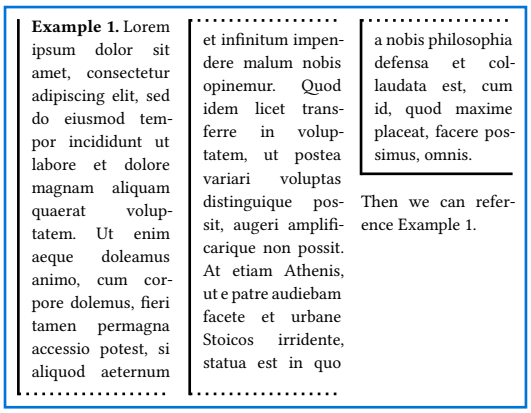
```
#show figure-where-kind-in(theofig-kinds): set
block(breakable: true)
// #show figure: set block(breakable: true) //
alternative

#set text(0.7em)
#show figure-where-kind-in(theofig-kinds):
block.with(
  stroke: 1pt, inset: 5pt, radius: 5pt,
)

#block(
  width: 100%, height: 150pt,
  stroke: 1pt + blue, inset: 5pt,
)[
  #columns(3)[
    #example[#lorem(60)]<example-1>

    Then we can reference @example-1.
  ]
]
```




Example with styled breakpoints for the blocks (see Breakpoints on broken blocks).

```
#show figure-where-kind-in(theofig-kinds): set
block(breakable: true)

#set text(0.6em)
#show figure-where-kind-in(
  theofig-kinds, except: "example"
): theofig-split-box
#show figure.where(kind: "example"): theofig-
split-box.with(
  stroke: (left: 1pt, bottom: 1pt, between: (dash:
"dotted")),
  inset: 5pt,
)

#block(
  width: 100%, height: 150pt,
  stroke: 1pt + blue, inset: 5pt,
)[
  #columns(3)[
    #theorem[#lorem(80)]<theorem-split-box>

    Then we can reference @theorem-split-box.
  ]
]
#block(
  width: 100%, height: 150pt,
  stroke: 1pt + blue, inset: 5pt,
)[
  #columns(3)[
    #example[#lorem(90)]<example-split-box>

    Then we can reference @example-split-box.
  ]
]
```



## Limitations

Because `theofig()` implemented as figure, show rules applied to it affect nested figures of any kind, including images and tables:

```
#show figure.where(kind: "example"): it => {
  set figure(numbering: "I")
  show figure.caption: smallcaps
  it
}
#example[
  Example with an image:
  #figure(caption: [Example Image])[
    #image(
      bytes(range(256).map(i => i.bit-and(i*i))),
      format:(encoding:"luma8",width:32,height:8),
      width: 100%,
    )
  ]
]
```

**Example I.** Example with an image:

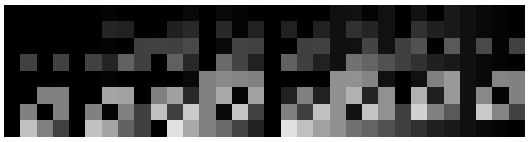


Figure I: Example Image

In this example, `smallcaps` is applied not only to the `example()` title, but also to the actual `figure`'s caption, the same is true for numbering. It is an undesirable limitation, which leads us to either moving our figures outside of `theofig()` environments or styling each `theofig()` environment individually like in the following example, using parameters `format-caption`, `format-body`, or `format-note`.

```
#let example = example.with(
  numbering: "I",
  format-caption: it => smallcaps(strong(it)),
)
#example[
  Example with an image:
  #figure(caption: [Example Image])[
    #image(
      bytes(range(256).map(i => i.bit-and(i*i))),
      format:(encoding:"luma8",width:32,height:8),
      width: 100%,
    )
  ]
]
```

**Example I.** Example with an image:

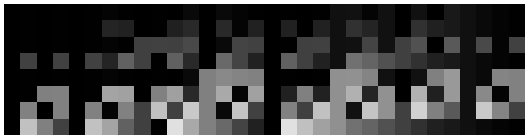


Figure 2: Example Image

# Main functions

## theofig()

This is the core factory function which implements a theorem-like environment on top of `figure`. Most user-facing environments (e.g. `#theorem`, `#definition`, `#proof`) are created with `.with(...)` specializations of this function:

```
#let definition  = theofig.with(kind: "definition",  supplement: "Definition",  translate-supplement: true)
#let theorem     = theofig.with(kind: "theorem",     supplement: "Theorem",     translate-supplement: true)
#let proof       = theofig.with(kind: "proof",       supplement: "Proof",       translate-supplement: true,
                                                                                numbering: none, qed: true)

#let lemma       = theofig.with(kind: "lemma",       supplement: "Lemma",       translate-supplement: true)
#let statement   = theofig.with(kind: "statement",   supplement: "Statement",   translate-supplement: true)
#let remark      = theofig.with(kind: "remark",      supplement: "Remark",      translate-supplement: true)
#let proposition = theofig.with(kind: "proposition", supplement: "Proposition", translate-supplement: true)
#let corollary   = theofig.with(kind: "corollary",   supplement: "Corollary",   translate-supplement: true,
                                                                                          numbering: none)

#let example     = theofig.with(kind: "example",     supplement: "Example",     translate-supplement: true)
#let algorithm   = theofig.with(kind: "algorithm",   supplement: "Algorithm",   translate-supplement: true)
#let problem     = theofig.with(kind: "problem",     supplement: "Problem",     translate-supplement: true)
#let solution    = theofig.with(kind: "solution",    supplement: "Solution",    translate-supplement: true,
                                                                                          numbering: none)
```

If you wish this list were extended, open an issue in repository, and it will probably added shortly.

```
theofig(
  body: content,
  kind: auto str,
  supplement: auto str content,
  number: auto int label other,
  numbering: auto none str function,
  block-options: dictionary,
  figure-options: dictionary,
  format-caption: none function array of functions,
  format-body: none function array of functions,
  format-note: none function array of functions,
  separator: none str content,
  translate-supplement: bool,
  qed: bool,
  ..note: array with one element,
)
```

---

**body** content

The contents of the environment.

### Example

```
#theorem[One body]
#theorem[Another body]
```

**Theorem 1.** One body

**Theorem 2.** Another body

---

**kind** auto or str default auto

The internal `figure.kind` value. If left `auto` and a string `supplement` is provided (not `auto` or `content`), the function will set `kind = lower(supplement)` so the kind can be inferred from the supplement like `"Theorem"` → `"theorem"`.

### Example

```
#let axiom = theofig.with(
  supplement: "Axiom", kind: "definition"
)
#definition[]
#axiom[
  Due to (`kind: "definition"`), numbering and
  show rules are shared with `#definition()`.
]
#definition[]
```

**Definition 1.**

**Axiom 2.** Due to (`kind: "definition"`), numbering and show rules are shared with `#definition()`.

**Definition 3.**

---

**`supplement`** `auto` or `str` or `content` default `auto`

The figure supplement (the textual label that appears as the environment title, e.g. `"Theorem"`, `"Definition"`). Behavior:

- `auto` — the function attempts to translate the `kind` using `theofig-translations` keyed by `text.lang` (so environments adapt to document language).
- If `supplement` explicitly provided as string and `translate-supplement` `==` `true`, the code will try to translate the supplement using `theofig-translations` dictionary using contextual `text.lang`. If there is no match in dictionary or `translate-supplement` `==` `false`, `supplement` is used as is.
- If `kind` is `auto` and `supplement`, `kind` is set to `lower(supplement)` (automatic kind from supplement).

### Example 1: Automatic kind selection

```
#let story = theofig.with(supplement: "Story")
#show figure.where(kind: "story"): text.with(
  fill: gradient.linear(
    red, orange, green, blue, fuchsia
  ),
)
#story[#lorem(25)]
#story[#lorem(5)]
```

**Story 1.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus.

**Story 2.** Lorem ipsum dolor sit amet.

### Example 2: Content supplement

```
#let dream = theofig.with(
  supplement: [D#sub[R]#super[E]#sub[A]M],
  kind: "dream",
)
#dream[I am in a @dream-2.]<dream-2>
```

**$D_R{}^E{}_A$M 1.** I am in a $D_R{}^E{}_A$M 1.

Note that if `kind` was omitted in the example 2, numeration would be shown, but counter would be 0 everywhere.

---

**`number`** `auto` or `int` or `label` or `other` default `auto`

Allows overriding the environment number. Behaviors:

- `auto` — default tracking (no manual override). `int` — uses that integer as the numbering (wrapped into a numbering function). If `numbering` `==` `none`, it produces an error.
- `label` — uses the counter number of an existing labeled figure; if `numbering` `==` `auto` the code also sets up `numbering` to produce the same numbering as the labeled figure's counter value.

- other values are used verbatim with `numbering = (..) => number`.

---

**`numbering`** `auto` or `none` or `str` or `function` default `auto`

The formatting function/style for the displayed number (e.g. Roman, alphabetic). Setting `auto` makes the function use either `figure.numbering` (default), otherwise this argument takes precedence over `figure.numbering`, which can be set using show rules or `figure-options` argument.

```
#definition[It is @def-r-1.]<def-r-1>

#show figure.where(kind: "definition"): set
figure(numbering: "A")
#definition[It is @def-r-2.]<def-r-2>

#let definition = definition.with(
  figure-options: (numbering: "I"),
)
#definition[It is @def-r-3.]<def-r-3>

#let definition = definition.with(
  numbering: "(i)",
)
#definition[It is @def-r-4.]<def-r-4>

#let definition = definition.with(
  numbering: numbering.with("(i)"),
)
#definition[It is @def-r-5.]<def-r-5>
```

**Definition 1.** It is Definition 1.

**Definition B.** It is Definition B.

**Definition III.** It is Definition III.

**Definition (iv).** It is Definition iv.

**Definition (v).** It is Definition (v).

---

**`block-options`** `dictionary` default `(:)`

Options passed to the inner `block(...)` call; use to control visual block styling (stroke, inset, radius, breakable, width etc.) without affecting nested blocks (as show rules do).

---

**`figure-options`** `dictionary` default `(:)`

Options passed to `figure(...)`. If `numbering` is determined (not `auto`), `figure-options` is augmented with `numbering: numbering`.

---

**`format-caption`** `none` or `function` or `array of functions` default `strong`

Function(s) applied to the `supplement` (the title part) before rendering. If `none`, no special formatting is applied. Typical values: `emph`, `smallcaps`, `strong`, or user-provided functions.

### Example

```
#theorem[]
#theorem(format-caption: none)[]
#theorem(format-caption: emph)[]
#theorem(format-caption:(smallcaps,underline))[]
```

**Theorem 1.**

Theorem 2.

*Theorem 3.*

Theorem 4.

**format-body** none or function or array of functions default none

Function(s) applied to the body content (the environment contents). If none, no additional formatting is applied.

### Example

```
#theorem[#lorem(5)]
#theorem(format-body: emph)[#lorem(5)]
#theorem(
  format-body: (text.with(blue), smallcaps,)
)[#lorem(4)]
```

**Theorem 1.** Lorem ipsum dolor sit amet.

**Theorem 2.** *Lorem ipsum dolor sit amet.*

**Theorem 3.** LOREM IPSUM DOLOR SIT.

**format-note** none or function or array of functions default it => [(#it)]

Function(s) applied to the note content (additional info to the caption, like authorship, date, or source). If none, no additional formatting is applied.

### Example

```
#theorem[Note][#lorem(3)]
#theorem(format-note: none)[`{Note}`][#lorem(3)]
#theorem(format-note:
  it => strong(delta: -300, [[#it]])
)[Note][#lorem(3)]
#theorem(format-note: (raw, emph), "note")
[#lorem(3)]
```

**Theorem 1 (Note).** Lorem ipsum dolor.

**Theorem 2 {Note}.** Lorem ipsum dolor.

**Theorem 3** [Note]**.** Lorem ipsum dolor.

**Theorem 4** ***note.*** Lorem ipsum dolor.

**separator** none or str or content default auto

Text appended between caption (supplement + caption + numbering) and the body. Behavior:

- auto — follows figure.caption.separator. If figure.caption.separator is also auto, separator becomes ".".
- Otherwise, this argument takes precedence over figure.caption.separator.

### Example

```
#definition[Default separator.]

#show figure.where(
  kind: "definition"
): set figure.caption(separator: "?")
#definition[Separator from show rule.]

#let definition = definition.with(
  separator: none
)
#definition[Separator from `.with()`
specialization.]

#definition(separator: ":")[Separator from
argument.]
```

**Definition 1.** Default separator.

**Definition 2?** Separator from show rule.

**Definition 3** Separator from .with() specialization.

**Definition 4:** Separator from argument.

**translate-supplement** bool default false

Whether a provided `supplement` should be passed through `theofig-translations` dictionary to allow localized titles. If `true` and `supplement` provided, package will attempt to map the lowercased supplement through `theofig-translations` for the current contextual `text.lang`. Note that if figure and a reference to it are in different languages, figure caption and reference supplement will have different languages as well.

It is `false` by default but is set to `true` for all user-facing environments `#algorithm()`, `#corollary()`, `#definition()`, `#example()`, `#lemma()`, `#problem()`, `#proof()`, `#proposition()`, `#remark()`, `#solution()`, `#statement()`, and `#theorem()`.

---

**`qed`** `bool` default `false`

If `true`, a `math.qed` marker (rendered as a box ∎) is added after the body. This is used for `proof` to append the end-of-proof marker. Note that `math.qed` symbol can be changed using `show math.qed: ...` rule.

---

**`..note`** `array with one element`

Additional text in supplement, i.e. author, year, or source of the theorem, like in **Theorem 1 (Cauchy, 1831).**

### Example

```
#theorem[1910][$1 + 1 = 2$.]<th-1>
#theorem[Newton–Leibniz][
  $ integral_a^b f'(x) dif x = f(b) - f(a). $
]
Note the absence of note in reference of @th-1.
```

**Theorem 1 (1910).** $1 + 1 = 2$.

**Theorem 2 (Newton–Leibniz).**

$$\int_a^b f'(x)\,\mathrm{d}x = f(b) - f(a).$$

Note the absence of note in reference of Theorem 1.

## show-figure-caption()

Shorthand for

```
#{
  it => {
    show figure.caption: function1
    show figure.caption: function2
    ...
    it
  }
}
```

for a provided list of functions.

**Example**

```
#show figure.where(
  kind: "theorem"
): show-figure-caption(
  emph, strong.with(delta: -300)
)

#lemma[Default style.]
#theorem[Custom style.]
```

**Lemma 1.** Default style.

*Theorem 1.* Custom style.

If no functions are provided, `show-figure-caption()` is an identity function that leaves its argument unaffected.

`show-figure-caption(..functions: array of functions) -> function`

---

**`..functions`** `array of functions`

List of functions that are applied to `figure.caption` by `show figure.caption: func`.

## figure-where-kind-in()

Selector for firgures with selected kinds, combined with `selector.or()`.

Use case: Apply a single style to many theorem-like environments in one `#show` rule, or to style all default theofig kinds while excluding some (e.g. exclude `"proof"`):

```
#show figure-where-kind-in(
  theofig-kinds, except: ("proof",)
): block.with(
    stroke: 1pt, radius: 3pt, inset: 5pt,
)
#definition[]
#theorem[]
#proof[]
```

**Definition 1.**

**Theorem 1.**

**Proof.** ∎

`figure-where-kind-in(kinds: array, except: array) -> selector`

---

**`kinds`** `array`

A list of figure kinds (e.g. `"theorem"`, `"definition"`, ...) to include in the selector.

**except** `array` default `()`

A list of figure kinds to exclude from the resulting selector.

## theofig-reset-counters()

For every `kind` in `kinds` that is not in `except`, perform:

```
#counter(figure.where(kind: kind)).update(0)
```

which sets that figure counter to `0` (so next created figure increments from 1). Useful to restart numbering for groups of environments (e.g., at the start of a chapter, or before a block of examples).

**Example**

```
#let corollary = corollary.with(numbering: "1")
#show figure.where(kind: "theorem"): it => {
  theofig-reset-counters(("corollary",))
  it
}

#theorem[]
#corollary[Follows Theorem 1.]
#corollary[Follows Theorem 1.]

#theorem[]
#corollary[Follows Theorem 2.]
#corollary[Follows Theorem 2.]
```

**Theorem 1.**

**Corollary 1.** Follows Theorem 1.

**Corollary 2.** Follows Theorem 1.

**Theorem 2.**

**Corollary 1.** Follows Theorem 2.

**Corollary 2.** Follows Theorem 2.

`theofig-reset-counters(kinds: array, except: array)`

**kinds** `array`

List of figure kinds whose counters are to reset.

**except** `array` default `()`

List of kinds to skip when resetting.

## algorithm()

```
#algorithm[Note][#lorem(3)]
```

**Algorithm 1 (Note).** Lorem ipsum dolor.

## corollary()

```
#corollary[Note][#lorem(3)]
```

**Corollary (Note).** Lorem ipsum dolor.

## definition()

```
#definition[Note][#lorem(3)]
```

**Definition 1 (Note).** Lorem ipsum dolor.

## example()

```
#example[Note][#lorem(3)]
```

**Example 1 (Note).** Lorem ipsum dolor.

## lemma()

```
#lemma[Note][#lorem(3)]
```

**Lemma 1 (Note).** Lorem ipsum dolor.

## problem()

```
#problem[Note][#lorem(3)]
```

**Problem 1 (Note).** Lorem ipsum dolor.

## proof()

```
#proof[Note][#lorem(3)]
```

**Proof (Note).** Lorem ipsum dolor. ∎

## proposition()

```
#proposition[Note][#lorem(3)]
```

**Proposition 1 (Note).** Lorem ipsum dolor.

## remark()

```
#remark[Note][#lorem(3)]
```

**Remark 1 (Note).** Lorem ipsum dolor.

## solution()

```
#solution[Note][#lorem(3)]
```

**Solution (Note).** Lorem ipsum dolor.

## statement()

```
#statement[Note][#lorem(3)]
```

**Statement 1 (Note).** Lorem ipsum dolor.

## theorem()

```
#theorem[Note][#lorem(3)]
```

**Theorem 1 (Note).** Lorem ipsum dolor.

## theofig-kinds

List of default kinds of environments defined by this package:

`"algorithm"`, `"corollary"`, `"definition"`, `"example"`, `"lemma"`, `"problem"`, `"proof"`, `"proposition"`, `"remark"`, `"solution"`, `"statement"`, and `"theorem"`.

The purpose to this variable is to be used together with selector `figure-where-kind-in()` for styling:

```
#show figure-where-kind-in(
  theofig-kinds
): block.with(
    stroke: 1pt, radius: 3pt, inset: 5pt,
)
#definition[]
#theorem[]
#proof[]
```

**Definition 1.**

**Theorem 1.**

**Proof.** ∎

## theofig-translations `dictionary of dictionaries`

List of translated supplements. Which is used by `theofig()` to localize default supplements depending on contextual `text.lang` if the `translate-supplement` option of `theofig()` is true.

```
List of supported languages:
#theofig-translations.keys().join(", ").

For example German translations are

#theofig-translations.at("de")
```

List of supported languages: cs, de, en, es, fr, it, ja, ko, pl, pt, ru, zh.

For example German translations are

```
(
  algorithm: "Algorithmus",
  corollary: "Korollar",
  definition: "Definition",
  example: "Beispiel",
  lemma: "Lemma",
  problem: "Aufgabe",
  proof: "Beweis",
  proposition: "Proposition",
  remark: "Bemerkung",
  solution: "Lösung",
  statement: "Aussage",
  theorem: "Satz",
)
```

## theofig-split-box()

Create a box for content which spans multiple pages/columns and has custom borders above and below the column-break.

Source: https://sitandr.github.io/typst-examples-book/book/typstonomicon/block_break.html#implementation-via-headers-footers-and-stated

```
theofig-split-box(
  body,
  stroke: auto stroke dictionary,
  inset: length dictionary,
  ..args: arguments,
)
```

---

**stroke** `auto` or `stroke` or `dictionary` default `auto`

How to stroke the borders

This can be:

- `auto` for a stroke of `1pt + black` for `top,bottom,left,right` borders (without `between` borders)

- any kind of stroke for `top,bottom,left,right`
- A dictionary describing the stroke for each side individually. The dictionary can contain the following keys in order of precedence:
  - `left`, `right`, `top`, `bottom`, `between-top`, `between-bottom`,
  - `x` for `left` and `right`, `y` for `top` and `bottom`, `between` for `between-top` and `between-bottom`,
  - `outer` for all sides except `between-top` and `between-bottom`,
  - `rest` for all sides

---

**`inset`** `length` or `dictionary` default `5pt`

How much to pad the content of the split-box.

This can be:

- a length, that sets the padding for all sides with non-0pt strokes
- A dictionary describing the padding for each side individually. The dictionary can contain the following keys in order of precedence:
  - `left`, `right`, `top`, `bottom`, `between-top`, `between-bottom`,
  - `x` for `left` and `right`, `y` for `top` and `bottom`, `between` for `between-top` and `between-bottom`,
  - `outer` for all sides except `between-top` and `between-bottom`,
  - `rest` for all sides

---

**`..args`** `arguments`

Any extra named args will be sent to the underlying grid when called.

This is useful for fill, align, etc.